Adjunction Phenomenon on Generalized Pairs

We work over an algebraically closed field of characteristic zero, like $\mathbb{C}$. All varieties are assumed to be normal unless otherwise stated.

1 Adjunctions for generalized pairs

Let $(X, B, \mathbf{M})$ be a generalized (sub-)pair, where X is a normal algebraic variety, B is an $\mathbb{R}$-divisor, $\mathbf{M}$ is an $\mathbb{R}$-b-Cartier b-divisor which is b-nef, and $K_X + B + M_X$ is $\mathbb{R}$-Cartier. A *clean log resolution* $p : W \to X$ is a log resolution of the pair (X, B) on which $\mathbf{M}$ descends and the trace M_W is a nef $\mathbb{R}$-Cartier divisor. We may choose compatible canonical divisors and write the discrepancy formula

$$K_W + B_W + M_W = p^*(K_X + B + M_X),$$

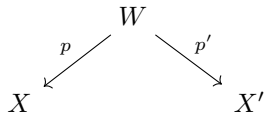
where B_W is also defined in this way.

Lemma 1. *Let $\rho : V \to W$ be a proper birational morphism of smooth varieties, where $\mathbf{M}$ descends to W. Then*

$$M_V = \rho^* M_W, \qquad K_V + B_V = \rho^*(K_W + B_W).$$

Proof. The first equality is the definition of descent. The other equality is trivial. See [b-Div]. □

Definition 2 (Crepant birational)**.** Let $(X, B, \mathbf{M})$ and $(X', B', \mathbf{M}')$ be generalized sub-pairs. We say that the two generalized sub-pairs are *crepant birational* if there is a common clean log resolution



such that, writing

$$K_W + B_W + M_W = p^*(K_X + B + M_X),$$
$$K_W + B'_W + M'_W = (p')^*(K_{X'} + B' + M'_{X'}),$$

we have

$$B_W = B'_W, \qquad M_W = M'_W.$$

Definition 3 (Generalized dlt)**.** A generalized sub-lc pair $(X, B, \mathbf{M})$ is *generalized (sub-)divisorial log terminal* if there is an open subset $U \subseteq X$ such that $(U, B|_U)$ is log smooth and every generalized lc centre T has its generic point $\eta_T \in U$, where $T \cap U$ is an lc centre of the pair $(U, B|_U)$.

Lemma 4. *Let $(X, B, \mathbf{M})$ be a generalized sub-pair with M_X $\mathbb{R}$-Cartier, and let $p : W \to X$ be a clean log resolution. Then*

$$F := p^* M_X - M_W \geq 0.$$

If B_W^{ord} *is defined by*

$$K_W + B_W^{\mathrm{ord}} = p^*(K_X + B),$$

then $B_W = B_W^{\mathrm{ord}} + F$.

In particular, if $(X, B, \mathbf{M})$ *is generalized sub-lc, then* (X, B) *is sub-lc and every lc place of* (X, B) *is a generalized lc place.*

Proof. F is p-exceptional by the definition of the b-divisor, and $-F$ is p-nef since M_W is nef. Thus the negativity lemma gives $F \geq 0$. The remaining statement comes from

$$A(E, X, B) = A(E, X, B, \mathbf{M}) + \mathrm{mult}_E F$$

for every prime divisor E on a clean resolution $W \to X$. □

Proposition 5. *If* $(X, B, \mathbf{M})$ *is a generalized dlt sub-pair with* X $\mathbb{Q}$*-factorial,*[1] *then* (X, B) *is sub-dlt and its lc centres are precisely the generalized lc centres.*

Proof. Let $U \subseteq X$ be an open subset such that $(U, B|_U)$ is log smooth and every generalized lc centre has its generic point in U. Take a log resolution $p : Y \to X$ which is an isomorphism over U. Every p-exceptional prime divisor $E \subseteq Y$ has $c_X(E) \subseteq X \setminus U$. If $A(E, X, B, \mathbf{M}) = 0$, then E is a generalized lc place and $c_X(E)$ is a generalized lc centre. This gives a contradiction. Hence

$$A(E, X, B) \geq A(E, X, B, \mathbf{M}) > 0$$

implies that (X, B) is sub-dlt.

Lemma 4 gives

$$A(E, X, B) = A(E, X, B, \mathbf{M}) + \mathrm{mult}_E F,$$

which says that each lc centre of (X, B) is a generalized lc centre. Conversely, for every generalized lc centre T with $\eta_T \in U$, we have

$$\overline{T \cap U} = T.$$

We may take a higher log resolution carrying a prime divisor E over U with $c_U(E) = T \cap U$ and $A(E, U, B|_U) = 0$. Let $\overline{E}$ be its closure on this model. At this time,

$$A(\overline{E}, X, B) = A(E, U, B|_U) = 0$$

gives that $T = c_X(\overline{E})$ is an lc centre of (X, B). □

Now, we introduce an important result in [Bir-Zhang14, Lemma 4.5], also proved in [HL22, Proposition 3.9].

Definition–Theorem 6 (Generalized dlt modifications)**.** Let $(X, B, \mathbf{M})$ be a quasi-projective generalized lc pair, with $\mathbf{M}$ b-nef and $\mathbb{R}$-b-Cartier. There is a projective birational morphism $r : X' \to X$ such that X' is $\mathbb{Q}$-factorial, $(X', B', \mathbf{M})$ is generalized dlt, and

$$K_{X'} + B' + M_{X'} = r^*(K_X + B + M_X).$$

Here

$$B' = \widetilde{B} + (r\text{-exceptional part}),$$

where $\widetilde{B}$ is the strict transform of B, with coefficients $0 < b_i \leq 1$. In particular, the r-exceptional part is reduced and has log discrepancy zero.

We call $(X', B', \mathbf{M})$ a *generalized dlt modification* of $(X, B, \mathbf{M})$.

[1] This condition ensures that $K_X + B$ is $\mathbb{R}$-Cartier, since $M_X = p_* M_W$ is a real Weil divisor.

For a proof, see [Bir-Zhang14, Lemma 4.5] or [HL22, Proposition 3.9].

Now, we recall that in [Har77, II, 8.20] we have the classical adjunction formula.

Lemma 7. *Let W be a smooth variety, with $S \subseteq W$ a smooth prime divisor. Then there is a natural isomorphism*

$$\omega_S \cong \big(\omega_W \otimes \mathcal{O}_W(S)\big)|_S,$$

or equivalently,

$$(K_W + S)|_S \sim K_S.$$

Restrictions of the nef part are taken using compatible Cartier representatives whose supports do not contain the divisor of restriction.

Proposition 8 (Adjunction on a clean log resolution)**.** *Let (W, B_W) be a log smooth sub-lc pair and let $S \subseteq W$ be a prime divisor. Suppose that $\operatorname{mult}_S B_W = 1$, and let M_W be a nef $\mathbb{R}$-Cartier divisor. Set*

$$B_S = (B_W - S)|_S, \qquad M_S = M_W|_S.$$

Then $(S, B_S, \overline{M_S})$ is generalized sub-lc, M_S is nef, and

$$(K_W + B_W + M_W)|_S \sim_{\mathbb{R}} K_S + B_S + M_S.$$

If $B_W \geq 0$, then $B_S \geq 0$, i.e., $(S, B_S, \overline{M_S})$ is generalized lc.

Proof. Write

$$B_W = S + \sum_i b_i D_i,$$

where each D_i is different from S. By the simple normal crossing property,

$$(B_W - S)|_S = \sum_i b_i D_i|_S$$

still has simple normal crossing support with $b_i \leq 1$. By [BGR, Corollary 5.2.20], we see that (S, B_S) is sub-lc. Since the b-divisor $\overline{M_S}$ descends to S itself, its nef terms can be cancelled when discrepancies are computed on higher models. Thus the generalized pair is sub-lc. The remaining assertions are obvious. □

2 Adjunction on the normalization

Let $(X, B, \mathbf{M})$ be a sub-lc generalized pair, and let S be a prime divisor with $\operatorname{mult}_S B = 1$. Here we do not assume that S is smooth or normal. Let $\nu : S^\nu \to S$ be its normalization, and let $i : S^\nu \to X$ be the normalization followed by the inclusion. Choose a clean log resolution $p : W \to X$, and let S_W be the strict transform of S. It is a smooth variety. Its morphism to S factors through S^ν by the universal property of the normalization, so there is a commutative diagram

$$\begin{array}{ccc} S_W & \xhookrightarrow{i_W} & W \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ S^\nu & \xrightarrow[i]{} & X. \end{array}$$

Here q is a proper birational morphism such that $i \circ q = p \circ i_W$, and $\operatorname{mult}_{S_W} B_W = 1$. We may regard $S_W \to S^\nu$ as a model of S^ν.

Definition 9 (Divisorial adjunction)**.** With notation as above, on S_W set

$$B_{S_W} = (B_W - S_W)|_{S_W}, \qquad M_{S_W} = M_W|_{S_W}.$$

The Cartier closure $\overline{M_{S_W}}$, regarded as a b-divisor over S^ν, is denoted by $\mathbf{M}^S$. Define its trace and the corresponding boundary divisor by

$$\operatorname{tr}_{S^\nu} \mathbf{M}^S = q_* M_{S_W}, \qquad B_{S^\nu} = q_* B_{S_W}.$$

The triple $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is called the generalized (sub-)pair obtained by *divisorial adjunction to* S^ν.

Lemma 10 (Crepant adjunction formula)**.** *The divisor $K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu}$ is $\mathbb{R}$-Cartier, and, with canonical divisors chosen compatibly,*

$$K_{S_W} + B_{S_W} + M_{S_W} = q^*\big(K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu}\big),$$

$$K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu} \sim_{\mathbb{R}} i^*(K_X + B + M_X).$$

Proof. The second relation need not be an equality because i is not dominant. Indeed, we have

$$\begin{aligned} K_{S_W} + B_{S_W} + M_{S_W} &\sim_{\mathbb{R}} i_W^*(K_W + B_W + M_W) \\ &= i_W^* p^*(K_X + B + M_X) \\ &= q^* i^*(K_X + B + M_X). \end{aligned}$$

We may choose compatible canonical divisors with $q_* K_{S_W} = K_{S^\nu}$, so that

$$q_*\big(K_{S_W} + B_{S_W} + M_{S_W}\big) = K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu}.$$

Pushing the displayed real linear equivalence forward gives

$$K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu} \sim_{\mathbb{R}} i^*(K_X + B + M_X),$$

so the divisor on the left is $\mathbb{R}$-Cartier. The difference between the two sides of the first asserted equality is q-exceptional and real linearly equivalent to zero. It is therefore zero by the negativity lemma, proving that equality.

Note that we often write $i^*(K_X + B + M_X)$ as $(K_X + B + M_X)|_{S^\nu}$. □

Proposition 11. *The construction of divisorial adjunction is independent of the chosen clean log resolution, with compatible representatives of canonical divisors. In particular, the b-divisor $\mathbf{M}^S$ is b-nef, and the induced generalized pair $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is sub-lc.*

Proof. Suppose that $r : W' \to W$ is a higher log resolution, let $S_{W'}$ be the strict transform of S_W, and let $r_S : S_{W'} \to S_W$ be the induced proper birational morphism. The commutative diagram

$$\begin{array}{ccc} S_{W'} & \hookrightarrow & W' \\ {\scriptstyle r_S}\downarrow & & \downarrow{\scriptstyle r} \\ S_W & \xhookrightarrow{i_W} & W \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ S^\nu & \xrightarrow[i]{} & X \end{array}$$

and descent of the b-divisor $\mathbf{M}$ give

$$M_{S_{W'}} = (r^* M_W)|_{S_{W'}} = r_S^*(M_W|_{S_W}) = r_S^* M_{S_W}.$$

Thus the Cartier closures of the two restricted nef divisors are the same b-divisor, that is,

$$\mathbf{M}^S = \overline{M_{S_W}} = \overline{r_S^* M_{S_W}},$$

and we have

$$\begin{aligned} \operatorname{tr}_{S^\nu} \mathbf{M}^S &= q_* M_{S_W} \\ &= q_*(r_S)_* r_S^* M_{S_W} \\ &= (q \circ r_S)_* M_{S_{W'}}. \end{aligned}$$

Applying adjunction to the equation

$$K_{W'} + B_{W'} = r^*(K_W + B_W)$$

gives

$$K_{S_{W'}} + B_{S_{W'}} = r_S^*(K_{S_W} + B_{S_W}),$$

where

$$B_{S_{W'}} = (B_{W'} - S_{W'})|_{S_{W'}}, \qquad B_{S_W} = (B_W - S_W)|_{S_W},$$

with compatible canonical divisors. Pushing forward to S^ν gives

$$\begin{aligned} (q \circ r_S)_*(K_{S_{W'}} + B_{S_{W'}}) &= q_*(r_S)_* r_S^*(K_{S_W} + B_{S_W}) \\ &= q_*(K_{S_W} + B_{S_W}), \end{aligned}$$

and hence the same boundary divisor

$$B_{S^\nu} = q_* B_{S_W} = (q \circ r_S)_* B_{S_{W'}}.$$

The nefness of M_{S_W} and $M_{S_{W'}}$ is obvious, so $\mathbf{M}^S$ is b-nef.

Finally, the log smooth pair (S_W, B_{S_W}) is sub-lc. Therefore, for any prime divisor E over S^ν which appears on a higher model $f : T \to S_W$,[2] we have $\operatorname{mult}_E B_T \leq 1$ for

$$K_T + B_T = f^*(K_{S_W} + B_{S_W}).$$

Since $\mathbf{M}^S$ descends to S_W, we have $M_T^S = f^* M_{S_W}^S$, and

$$\begin{aligned} K_T + B_T + M_T^S &= f^*(K_{S_W} + B_{S_W} + M_{S_W}^S) \\ &= (q \circ f)^*(K_{S^\nu} + B_{S^\nu} + M_{S^\nu}^S) \end{aligned}$$

by Lemma 10. Hence

$$A(E, S^\nu, B_{S^\nu}, \mathbf{M}^S) = 1 - \operatorname{mult}_E B_T \geq 0,$$

and we see that $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is sub-lc. □

[2]For any prime divisor E over S^ν, there is a normal model $Y \to S^\nu$ such that $E \subset Y$. Take a resolution $T \to W_0$, where W_0 is the irreducible component of $Y \times_{S^\nu} S_W$ which dominates S^ν, to obtain a commutative diagram

$$\begin{array}{ccc} T & \xrightarrow{w} & Y \\ {\scriptstyle f}\downarrow & & \downarrow \\ S_W & \longrightarrow & S^\nu. \end{array}$$

Here B_Y is the corresponding crepant boundary on Y. The morphism w is an isomorphism at the generic point of E, so that $\operatorname{mult}_E B_T = \operatorname{mult}_E B_Y$.

Definition 12 (Different)**.** Let V be a normal algebraic variety, let $S \subset V$ be a prime divisor, and let G be an $\mathbb{R}$-divisor not containing S as a component. Suppose that $K_V + S + G$ is $\mathbb{R}$-Cartier, and denote by $\nu : S^\nu \to S$ the normalization. Take a log resolution $p : W \to V$, let S_W be the strict transform of S, and let $q : S_W \to S^\nu$ be the induced morphism. Choose compatible canonical divisors and write

$$K_W + B_W^{\mathrm{ord}} = p^*(K_V + S + G),$$

where $\operatorname{mult}_{S_W} B_W^{\mathrm{ord}} = 1$. Define

$$\operatorname{Diff}_{S^\nu}(G) = q_*\big((B_W^{\mathrm{ord}} - S_W)|_{S_W}\big),$$

which is called the *different of G on S^ν*.

Remark. The above definition is independent of the choice of log resolution $W \to V$. This is easy to verify: use the independence argument of Proposition 11 with $\mathbf{M} = 0$. Note that when $\mathbf{M} = 0$, the divisor B_{S^ν} in divisorial adjunction coincides with $\operatorname{Diff}_{S^\nu}(G)$. Thus Lemma 10 gives

$$(K_V + S + G)|_{S^\nu} \sim_{\mathbb{R}} K_{S^\nu} + \operatorname{Diff}_{S^\nu}(G).$$

Proposition 13 (Adjunction inequality)**.** *Let $(V, S + G, \mathbf{P})$ be an lc generalized pair, where V is $\mathbb{Q}$-factorial, $S \subset V$ is a prime divisor, and $G \geq 0$ does not contain S. The crepant adjunction formula gives*

$$(K_V + S + G + P_V)|_{S^\nu} \sim_{\mathbb{R}} K_{S^\nu} + G_{S^\nu} + P^S_{S^\nu},$$

where

$$G_{S^\nu} = q_*\big((B_W - S_W)|_{S_W}\big)$$

is the boundary divisor in divisorial adjunction. Then

$$G_{S^\nu} \geq G|_{S^\nu},$$

where the right-hand side is well-defined since V is $\mathbb{Q}$-factorial and $S \not\subseteq \operatorname{Supp} G$.

Proof. Take a log resolution $p : W \to V$ on which $\mathbf{P}$ descends. Denote the strict transform of S by S_W, with induced proper birational morphism $q : S_W \to S^\nu$. Set

$$F = p^* P_V - P_W.$$

It is exceptional and $-F$ is nef over V, so $F \geq 0$ by the negativity lemma. Since $p_* F = 0$, we have $S_W \not\subseteq \operatorname{Supp} F$, and hence $F|_{S_W} \geq 0$.

Let B_W^{ord} be the crepant boundary of the ordinary pair $(V, S + G)$ on W. Then the generalized crepant boundary

$$B_W = B_W^{\mathrm{ord}} + F$$

gives

$$\begin{aligned} G_{S^\nu} &= q_*\big((B_W - S_W)|_{S_W}\big) \\ &= \operatorname{Diff}_{S^\nu}(G) + q_*(F|_{S_W}), \end{aligned}$$

where the second summand is effective. Moreover,

$$\begin{aligned} \operatorname{Diff}_{S^\nu}(0) &= q_*\big((B_W^{\mathrm{ord}} - p^*G - S_W)|_{S_W}\big) \\ &= \operatorname{Diff}_{S^\nu}(G) - G|_{S^\nu} \\ &\geq 0; \end{aligned}$$

see [Kol13, Proposition 4.5]. Thus $G_{S^\nu} \geq G|_{S^\nu}$. □

Theorem 14 (Generalized dlt adjunction)**.** *Let $(X, B, \mathbf{M})$ be a dlt generalized pair, and let S be a component of $\lfloor B \rfloor$. Divisorial adjunction on the normalization $\nu : S^\nu \to S$ gives a dlt generalized pair $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ with $B_{S^\nu} \geq 0$. Its lc centres map under ν precisely onto those lc centres of $(X, B, \mathbf{M})$ that are properly contained in S.*

Proof. Step 1. Proposition 11 shows that $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is sub-lc. Since effectivity, the dlt condition and the stated correspondence of lc centres can be checked locally on X, we may assume that X is affine. Take a generalized dlt modification $r : X' \to X$, and denote by S' the strict transform of S. Since X' is $\mathbb{Q}$-factorial, Proposition 5 shows that (X', B') is a dlt pair, where B' is defined by

$$K_{X'} + B' = r^*(K_X + B + M_X) - M_{X'}.$$

Moreover, $\operatorname{mult}_S \lfloor B \rfloor = 1$ gives $\operatorname{mult}_S B \geq 1$, and log canonical singularities give

$$0 \leq A(S, X, B, \mathbf{M}) = 1 - \operatorname{mult}_S B.$$

Thus

$$\operatorname{mult}_S B = \operatorname{mult}_{S'} B' = 1.$$

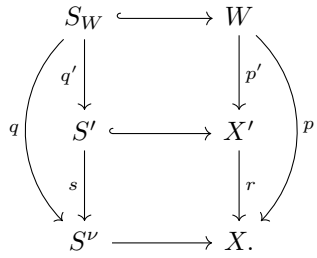
By [KM98, Corollary 5.52], S' is normal. Using [BGR, Proposition 5.2.43] and [BGR, Lemma 5.2.11] twice, the pair $(X', 0)$ is klt. Applying Proposition 13 to $(X', B', \mathbf{M})$ gives

$$B_{S'} \geq (B' - S')|_{S'} \geq 0,$$

where

$$B_{S'} = q'_*\big((B_W - S_W)|_{S_W}\big)$$

for a common higher data log resolution $W \to X', X$ on which $\mathbf{M}$ descends, with induced projective birational morphism $q' : S_W \to S'$. The induced morphism $s : S' \to S^\nu$ is projective birational. The following diagram commutes:



We have

$$\begin{aligned} B_{S^\nu} &= q_*\big((B_W - S_W)|_{S_W}\big) \\ &= (s \circ q')_*\big((B_W - S_W)|_{S_W}\big) \\ &= s_* B_{S'} \geq 0. \end{aligned}$$

Step 2. Now choose the open subset $U \subseteq X$ witnessing generalized dlt. Since $S \cap U$ is smooth, its inverse image in S^ν is isomorphic to $S \cap U$, and we identify them for convenience. Applying Proposition 13 to the log smooth pair $(U, B|_U, \mathbf{M})$ gives

$$B_{S^\nu}|_{S \cap U} \geq (B - S)|_{S \cap U}.$$

The ordinary pair $(S \cap U, B_{S^\nu}|_{S \cap U})$ is lc by Lemma 4, and $S \cap U$ is $\mathbb{Q}$-factorial. Since $(S \cap U, (B - S)|_{S \cap U})$ is log smooth, its lc centres are exactly the irreducible components

of non-empty intersections of the coefficient-one components of $(B-S)|_{S\cap U}$. Let E be an arbitrary lc place obtained by blowing up an lc centre of $(S\cap U,(B-S)|_{S\cap U})$, or the centre itself if it is divisorial. Write $a: Y\to S\cap U$ for the corresponding birational morphism. Then

$$\begin{aligned}0 &= A\big(E, S\cap U, (B-S)|_{S\cap U}\big)\\ &= A\big(E, S\cap U, B_{S^\nu}|_{S\cap U}\big)\\ &\quad + \operatorname{mult}_E a^*\Big(B_{S^\nu}|_{S\cap U} - (B-S)|_{S\cap U}\Big).\end{aligned}$$

Every summand is non-negative, so

$$A\big(E, S\cap U, B_{S^\nu}|_{S\cap U}\big) = 0.$$

Now, for any lc centre $T\subsetneq S$ of the generalized dlt pair $(X,B,\mathbf{M})$, its generic point η_T lies in U by definition, and $T\cap U$ is an lc centre of the log smooth pair $(U, B|_U)$. Hence $T\cap U$ is an lc centre of $(S\cap U,(B-S)|_{S\cap U})$. Choose an lc place E whose centre is $T\cap U$. The preceding analysis and Lemma 4 give

$$\begin{aligned}0 &\le A(E, S^\nu, B_{S^\nu}, \mathbf{M}^S)\\ &= A\big(E, S\cap U, B_{S^\nu}|_{S\cap U}, \mathbf{M}^S\big)\\ &\le A\big(E, S\cap U, B_{S^\nu}|_{S\cap U}\big) = 0.\end{aligned}$$

Thus E is also a generalized lc place of $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$, and

$$C = \overline{\nu^{-1}(T\cap U)} \subseteq S^\nu$$

is an lc centre of this generalized pair.

Conversely, let $C\subseteq S^\nu$ be an lc centre of $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$. On a higher data log resolution $q: S_W\to S^\nu$, where $\mathbf{M}^S$ descends, we have

$$\begin{aligned}A(E, S_W, B_{S_W}) &= A(E, S_W, B_{S_W}, \mathbf{M}^S)\\ &= A(E, S^\nu, B_{S^\nu}, \mathbf{M}^S) = 0,\end{aligned}$$

where E is an lc place with centre C. Thus the centre R of E on the log smooth pair (S_W, B_{S_W}) is an irreducible component of an intersection of coefficient-one components of B_{S_W}, and $C = q(R)$. But the coefficient-one components of

$$B_{S_W} = (B_W - S_W)|_{S_W}$$

come from coefficient-one components of B_W. Hence R is a component of

$$S_W\cap D_1\cap\cdots\cap D_r, \qquad \operatorname{mult}_{S_W} B_W = \operatorname{mult}_{D_j} B_W = 1.$$

Thus R is also an lc stratum of (W, B_W). Therefore

$$\nu(C) = p(R)$$

is an lc centre of $(X,B,\mathbf{M})$. This proves the second statement.

Step 3. Finally, consider the open subset of S^ν

$$S^0 = (S \cap U) \setminus \mathrm{Supp}\big(B_{S^\nu}|_{S\cap U} - (B-S)|_{S\cap U}\big),$$

on which

$$B_{S^\nu}|_{S^0} = (B-S)|_{S^0}$$

has snc support. For any lc centre $C \subseteq S^\nu$ of $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$, we have shown that

$$T = \nu(C) \subsetneq S$$

is an lc centre of the generalized dlt pair $(X, B, \mathbf{M})$. Thus $\eta_T \in S \cap U$, and $T \cap U$ is an lc centre of $(S \cap U, (B-S)|_{S\cap U})$. Our preceding calculation

$$\mathrm{mult}_E\, a^*\Big(B_{S^\nu}|_{S\cap U} - (B-S)|_{S\cap U}\Big) = 0,$$

where E is an lc place with centre $T \cap U$ on a model $a : Y \to S \cap U$, shows that $\eta_T \in S^0$. Since ν is an isomorphism over $S \cap U$, we have $\eta_C \in S^0$. This proves that $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is generalized dlt. □

Corollary 15. *Let $(X, B, \mathbf{M})$ be an lc generalized pair, and let S be a component of $\lfloor B \rfloor$. Its divisorial adjunction $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is generalized lc.*

Proof. We only need to prove that $B_{S^\nu} \geq 0$, and this is the same as Step 1 of the preceding theorem. □

3 Adjunction on lc strata

Recall that an *lc stratum* means either the ambient variety or an lc centre. Let $(X, B, \mathbf{M})$ be $\mathbb{Q}$-factorial generalized dlt. For an lc stratum $\bar{T}$, we choose a chain of divisorial strata

$$X = \bar{T}_0 \supseteq \bar{T}_1 \supseteq \cdots \supseteq \bar{T}_c = \bar{T}, \qquad \mathrm{codim}_{\bar{T}_{i-1}} \bar{T}_i = 1.$$

Each $\bar{T}_i$ is normal, since it is an lc stratum of the dlt pair (X, B) by Proposition 5, and normality comes from [Kol13, Theorem 4.16]. Successive divisorial adjunction gives a dlt generalized pair $(\bar{T}_i, B_{\bar{T}_i}, \mathbf{M}^{(i)})$ at each step, by Theorem 14; we also write $\mathbf{M}^{\bar{T}_i}$ for this nef part. We may choose a common resolution

$$\begin{array}{ccccccccc} W = R_0 & \supseteq & R_1 & \supseteq & \cdots & \supseteq & R_c = R \\ {\scriptstyle p}\big\downarrow & & & & & & \\ X = \bar{T}_0 & \supseteq & \bar{T}_1 & \supseteq & \cdots & \supseteq & \bar{T}_c = \bar{T} \end{array}$$

such that each $R_i \to \bar{T}_i$ is a proper birational morphism and $\mathbf{M}$ descends to W, while its nef data M_W, restricted successively, gives the traces on each R_i.

Lemma 16 (Construction)**.** *For the above chosen chain, one can take a sufficiently high model $p : W \to X$ on which $\mathbf{M}$ descends, with an lc stratum $R \subseteq W$ mapping birationally onto $\bar{T}$, and such that $\mathbf{M}^{\bar{T}}$ descends to R and its trace there is $M_W|_R$.*

Proof. For the normal prime divisor $\bar{T}_1$, which is an lc centre of $(X, B, \mathbf{M})$, we have

$$0 = A(\bar{T}_1, X, B, \mathbf{M}) = 1 - \operatorname{mult}_{\bar{T}_1} B,$$

so $\operatorname{mult}_{\bar{T}_1} B = 1$, and we can take divisorial adjunction for $\bar{T}_1$. Taking its strict transform R_1 on W, with the following diagram

$$\begin{array}{ccc} R_1 & \hookrightarrow & W \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ \bar{T}_1 & \hookrightarrow & X, \end{array}$$

the construction of divisorial adjunction determines

$$\begin{aligned} B_{R_1} &= (B_W - R_1)|_{R_1}, \quad & M_{R_1} &= M_W|_{R_1}, \\ \mathbf{M}^{\bar{T}_1} &= \overline{M_{R_1}}, & B_{\bar{T}_1} &= q_* B_{R_1}. \end{aligned}$$

Note that for a proper birational morphism $R_1 \to \bar{T}_1$ to a normal algebraic variety, the image of its exceptional locus has codimension at least two in $\bar{T}_1$, so it is an isomorphism over codimension-one points. Thus the divisor $\bar{T}_2 \subseteq \bar{T}_1$ has a strict transform $R_2 \subseteq R_1$ which maps birationally onto $\bar{T}_2$. Theorem 14 says that $\bar{T}_2$ is an lc centre of the induced dlt generalized pair $(\bar{T}_1, B_{\bar{T}_1}, \mathbf{M}^{\bar{T}_1})$, that is,

$$0 = A(\bar{T}_2, \bar{T}_1, B_{\bar{T}_1}, \mathbf{M}^{\bar{T}_1}) = 1 - \operatorname{mult}_{\bar{T}_2} B_{\bar{T}_1}.$$

Hence $\operatorname{mult}_{\bar{T}_2} B_{\bar{T}_1} = 1$, and we can take divisorial adjunction to $\bar{T}_2$. On a further resolution, if necessary, the nef data restrict as

$$(M_W|_{R_1})|_{R_2} = M_W|_{R_2} = M_{R_2}.$$

Repeat this process for the finite chain; at each step the generalized dlt adjunction theorem gives a dlt generalized pair $(\bar{T}_i, B_{\bar{T}_i}, \mathbf{M}^{\bar{T}_i})$. □

Corollary 17 (Abundance on lc strata)**.** *Let $(X, B, \mathbf{M})$ be an lc generalized pair with X projective. Suppose that $\mathbf{M}$ is b-log abundant with respect to $(X, B, \mathbf{M})$; see [b-Div]. Let $r : X' \to X$ be a generalized dlt modification such that $(X', B', \mathbf{M})$ is generalized dlt and X' is $\mathbb{Q}$-factorial. Then, for every lc stratum $\bar{T} \subseteq X'$, if $(\bar{T}, B_{\bar{T}}, \mathbf{M}^{\bar{T}})$ is obtained by successive divisorial adjunction, the b-divisor $\mathbf{M}^{\bar{T}}$ is b-abundant.*

Proof. It follows from Lemma 16 that we may choose a sufficiently high common resolution

$$W \xrightarrow{\ p\ } X' \xrightarrow{\ r\ } X$$

on which $\mathbf{M}$ descends, and an lc stratum $R \subseteq W$ mapping birationally onto $\bar{T} \subseteq X'$ such that $\mathbf{M}^{\bar{T}}$ descends to R, with

$$\operatorname{tr}_R \mathbf{M}^{\bar{T}} = M_W|_R.$$

Since $\mathbf{M}$ is b-log abundant, $M_W|_R$ is abundant for every such lc stratum $R \subseteq W$. Then

$$M_R^{\bar{T}} = M_W|_R$$

implies that $\mathbf{M}^{\bar{T}}$ is b-abundant. For $\bar{T} = X'$, take $R = W$. □

Now, we give an application of Theorem 14.

Corollary 18. *Let $f : X \to Y$ be a proper morphism, and let $(X, B, \mathbf{M})$ be a $\mathbb{Q}$-factorial dlt generalized pair. Let $S \subseteq X$ be an lc stratum, and put $\bar{T} = f(S)$. Assume that $f(R) \subsetneq \bar{T}$ for every lc centre R of $(X, B, \mathbf{M})$ with $R \subsetneq S$. Let $(S, B_S, \mathbf{M}^S)$ be obtained by successive divisorial adjunction. Then it is generalized klt over general points of $\bar{T}$, that is, there is a non-empty open subset $\bar{T}^\circ \subseteq \bar{T}$ such that $(S, B_S, \mathbf{M}^S)$ is generalized klt on $(f|_S)^{-1}(\bar{T}^\circ)$.*

Proof. By Theorem 14, applied successively, every lc centre of $(S, B_S, \mathbf{M}^S)$ is an lc centre of $(X, B, \mathbf{M})$ properly contained in S. By our assumption, none of them dominates $\bar{T}$. Since $(S, B_S, \mathbf{M}^S)$ is generalized lc, we may choose

$$\bar{T}^\circ = \bar{T} \setminus \bigcup_R f(R),$$

where R runs over the finitely many lc centres of $(S, B_S, \mathbf{M}^S)$. Their images are proper closed subsets of $\bar{T}$, since f is proper. Thus $\bar{T}^\circ$ is an open dense subset over which $(S, B_S, \mathbf{M}^S)$ is generalized klt. □

References

[b-Div] *b-Divisors and Abundance for Generalized Pairs*, notes.

[BGR] *Birational Geometry Seminar: Chapters 1–6 and Appendices A–D. Revised Reading Notes.*

[Bir-Zhang14] C. Birkar and D.-Q. Zhang, *Effectivity of Iitaka fibrations and pluricanonical systems of polarized pairs*, Publications Mathématiques de l'IHÉS **123** (2016), 283–331.

[Har77] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics **52**, Springer, 1977.

[HL22] J. Han and Z. Li, *Weak Zariski decompositions and log terminal models for generalized pairs*, Mathematische Zeitschrift **302** (2022), 707–741.

[KM98] J. Kollár and S. Mori, *Birational Geometry of Algebraic Varieties*, Cambridge Tracts in Mathematics **134**, Cambridge University Press, 1998.

[Kol13] J. Kollár, with the collaboration of S. Kovács, *Singularities of the Minimal Model Program*, Cambridge Tracts in Mathematics **200**, Cambridge University Press, 2013.